

# 3.1 Newton's Laws of Motion

## Question Paper

|            |                             |
|------------|-----------------------------|
| Course     | CIEA Level Physics          |
| Section    | 3. Dynamics                 |
| Topic      | 3.1 Newton's Laws of Motion |
| Difficulty | Hard                        |

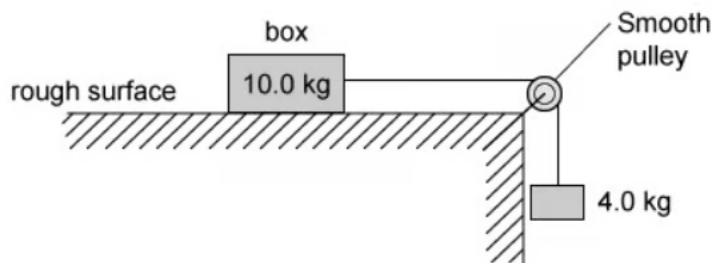
**Time allowed:** 20

**Score:** /10

**Percentage:** /100

**Question 1**

A box of mass  $10.0\text{ kg}$  rests on a horizontal rough surface. A string attached to the box passes over a smooth pulley and supports a  $4.0\text{ kg}$  mass at its other end.



When the box is released, a frictional force of  $12.0\text{ N}$  acts on it.

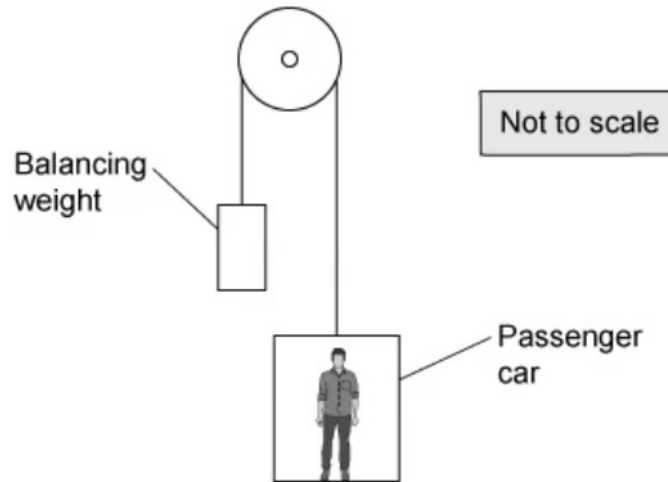
What is the acceleration of the box?

- A.  $1.2\text{ m s}^{-2}$
- B.  $1.9\text{ m s}^{-2}$
- C.  $2.7\text{ m s}^{-2}$
- D.  $3.0\text{ m s}^{-2}$

[1 mark]

## Question 2

A lift consists of a passenger car supported by a cable which runs over a light, frictionless pulley to a balancing weight. The balancing weight falls as the passenger car rises.



Some masses are shown in the table.

|                  | Mass / kg |
|------------------|-----------|
| passenger car    | 1200      |
| balancing weight | 1320      |
| passenger        | 80        |

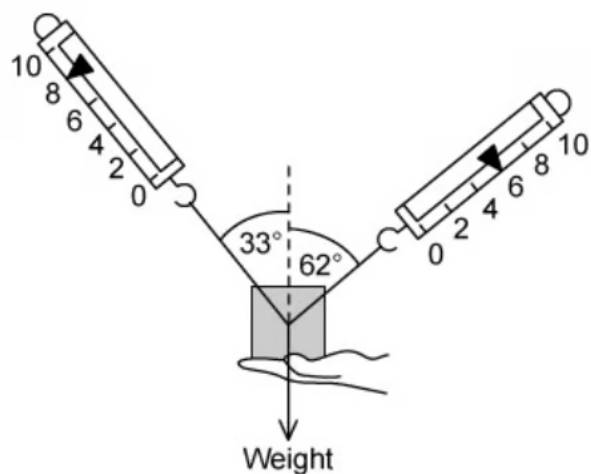
What is the magnitude of the acceleration of the car when carrying just one passenger and when the pulley is free to rotate?

- A.  $0.15 \text{ m s}^{-2}$
- B.  $0.30 \text{ m s}^{-2}$
- C.  $0.45 \text{ m s}^{-2}$
- D.  $0.92 \text{ m s}^{-2}$

[1 mark]

### Question 3

A 3.2 kg mass is supported by a person's hand and two newton-meters as shown.



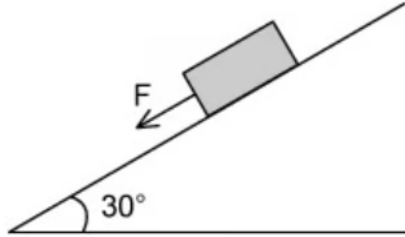
When the person's hand is removed, what is the initial vertical acceleration of the mass?

- A.  $3.0 \text{ m s}^{-2}$
- B.  $5.4 \text{ m s}^{-2}$
- C.  $6.8 \text{ m s}^{-2}$
- D.  $12.8 \text{ m s}^{-2}$

[1 mark]

**Question 4**

A brick of mass  $1.8\text{ kg}$  rests on a plane inclined at an angle of  $30^\circ$ . The weight of the brick has a component of  $F\text{ N}$  parallel with the plane. The brick also experiences a frictional force of  $3.0\text{ N}$



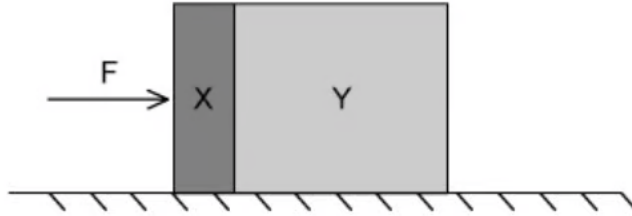
What is the acceleration of the brick down the plane?

- A.  $3.2\text{ m s}^{-2}$
- B.  $3.9\text{ m s}^{-2}$
- C.  $6.6\text{ m s}^{-2}$
- D.  $8.1\text{ m s}^{-2}$

[1 mark]

**Question 5**

Two blocks X and Y, of masses  $m$  and  $4m$  respectively, are accelerated along a smooth horizontal surface by a force  $F$  applied to block X as shown



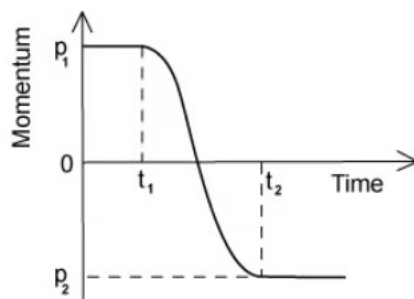
What is the magnitude of the force exerted on block Y by block X during this acceleration?

- A.  $\frac{F}{4}$
- B.  $\frac{F}{5}$
- C.  $\frac{4F}{5}$
- D.  $\frac{5F}{4}$

[1 mark]

## Question 6

The graph shows the variation with time of the momentum of a ball as it is kicked in a straight line.



Initially, the momentum is  $p_1$  at time  $t_1$ . At time  $t_2$  the momentum is  $p_2$ .

What is the magnitude of the average force acting on the ball between times  $t_1$  and  $t_2$ ?

A.  $\frac{p_1 - p_2}{t_1 - t_2}$

B.  $\frac{p_2 - p_1}{t_2 - t_1}$

C.  $\frac{p_2 - p_1}{t_1 - t_2}$

D.  $\frac{p_1 - p_2}{t_2 - t_1}$

[1 mark]

**Question 7**

The three forces acting on a hot-air balloon that is moving vertically are its weight, the force due to air resistance and the upthrust force.

The hot air balloon descends vertically at constant speed. The force of air resistance on the balloon is  $F$ .

Which weight of material must be released from the balloon so that it ascends vertically at the same constant speed?

- A.  $\frac{1}{2}F$
- B.  $F$
- C.  $2F$
- D.  $4F$

**[1 mark]****Question 8**

A ship of mass  $6.7 \times 10^7$  kg is approaching a harbour with speed  $18.9 \text{ m s}^{-1}$

By using reverse thrust it can maintain a constant total stopping force of 880,000 N

How long will it take to stop?

- A. 14 seconds
- B. 140 seconds
- C. 24 minutes
- D. 240 minutes

**[1 mark]**

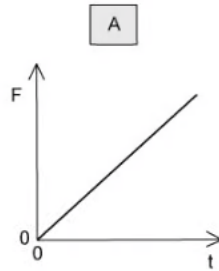


**Question 9**

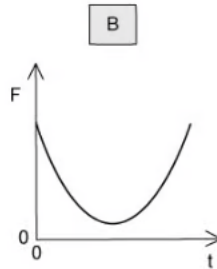
A tennis ball is dropped onto a table and bounces back up. The table exerts a force  $F$  on the ball.

Which graph best shows the variation with time  $t$  of the force  $F$  while the ball is in contact with the table?

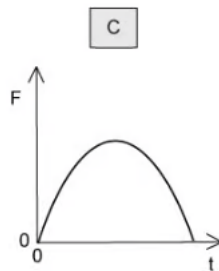
A.



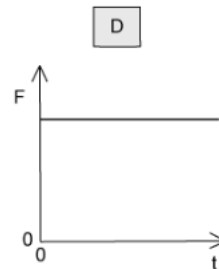
B.



C.



D.



[1 mark]



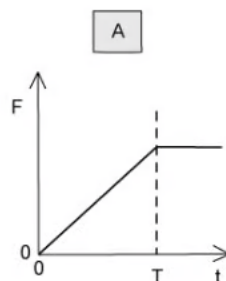
## Question 10

A skydiver falls vertically from a stationary balloon.

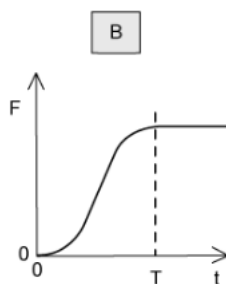
She leaves the balloon at time  $t = 0$ . At time  $t = T$ , she reaches terminal velocity. Beyond the time shown in the graphs, she opens her parachute

Which graph shows the variation with time  $t$  of the force  $F$  due to air resistance?

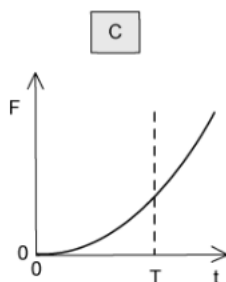
A.



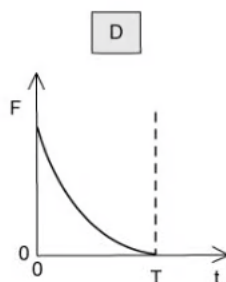
B.



C.



D.



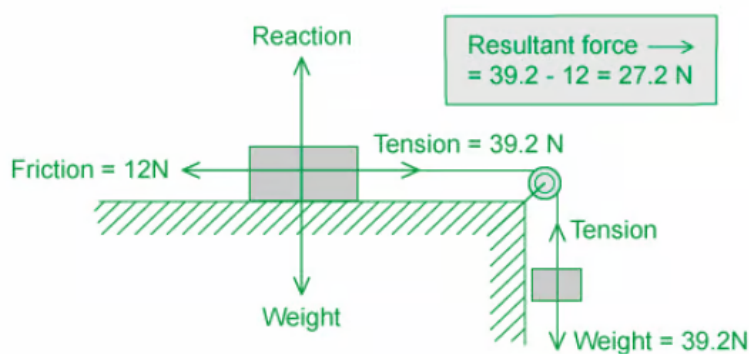
[1 mark]

## Model Answers: Hard

1

The correct answer is **B** because:

- The weight of the 4.0 kg mass pulls the 10.0 kg box to the right with a force equal to its weight
  - $\text{Weight} = mg = 4.0 \times 9.8 = 39.2 \text{ N}$
- The frictional force of 12.0 N opposes the motion of the 10.0 kg box, so it acts to the left
  - The resultant force is thus equal to  $39.2 - 12.0 = 27.2 \text{ N}$
- It is important to remember that the mass and the box move as a system so their masses must be added together
- Total mass of the system  $= 10.0 + 4.0 = 14.0 \text{ kg}$ 
  - Resultant force,  $F = ma$
  - $a = \frac{F}{m} = \frac{27.2}{14} = 1.9 \text{ m s}^{-2}$



2

The correct answer is **A** because:

- The weight of the car and passenger causes the lift to move downwards, while the

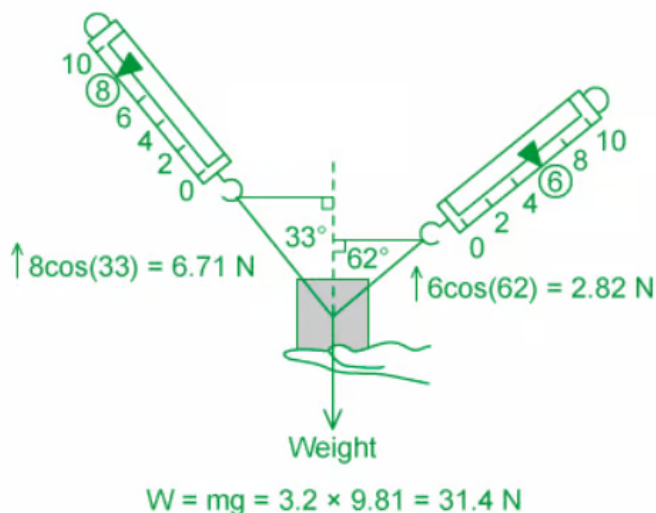
weight of the balancing weight causes the lift to move upwards

- Resultant (upward) force on car,  $F = 1320\text{ g} - 1200\text{ g} - 80\text{ g} = 40\text{ g}$ 
  - where  $g = 9.81\text{ N kg}^{-1}$
- $F = 40 \times 9.81 = 392\text{ N}$
- This resultant force is responsible for the acceleration of the passenger car, the balancing weight and the passenger, so we must use the total mass of the system, not just the mass of the lift and passenger
  - Total mass,  $m = 1320 + 1200 + 80 = 2600\text{ kg}$
- $F = ma$  rearranges to
  - $a = \frac{F}{m} = \frac{392}{2600} = 0.15\text{ m s}^{-2}$

3

The correct answer is **C** because:

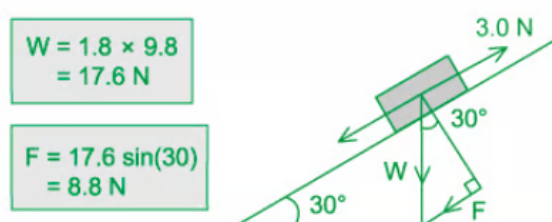
- We are looking for the vertical acceleration, so the horizontal components of force can be ignored
- The resultant upward force is equal to
  - $8 \cos 33 + 6 \cos 62 = 6.71 + 2.82 = 9.53\text{ N}$
- The resultant downward force is equal to the weight of the block
  - $W = mg = 3.2 \times 9.81 = 31.4\text{ N}$
- The overall resultant force is equal to
  - $F = 31.4 - 9.53 = 21.9\text{ N}$
  - Acceleration =  $\frac{F}{m} = \frac{21.9}{3.2} = 6.8\text{ m s}^{-2}$



4

The correct answer is **A** because:

- Solving problems on a plane is similar to solving problems vertically and horizontally, but instead, we resolve the forces parallel and perpendicular to the plane
- Since we are trying to find the acceleration of the block, we need to resolve the forces parallel to the plane
  - Weight =  $mg = 1.8 \times 9.8 = 17.6 \text{ N}$
  - Resultant force (down the plane) =  $W \sin 30 = 17.6 \sin 30 = 8.8 \text{ N}$
- The frictional force opposes the motion of the brick, so it acts up the plane
- So, the resultant force on the brick =  $8.8 - 3.0 = 5.8 \text{ N}$ 
  - Acceleration of the brick,  $a = \frac{F}{m} = \frac{5.8}{1.8} = 3.2 \text{ m s}^{-2}$



5

The correct answer is **C** because:

- The force  $F$  is applied to block X, which in turn applies a force to block Y
- As a result, both blocks accelerate
- Total mass being accelerated =  $m + 4m = 5m$
- The acceleration of the blocks is therefore equal to
  - $a = \frac{F}{5m}$
- We can use Newton's Second Law again to calculate the force on block Y only
  - $F_y = (4m)a = (4m)\left(\frac{F}{5m}\right) = \frac{4F}{5}$

6

The correct answer is **D** because:

- Force is equal to the rate of change of momentum
  - $F = \frac{\Delta p}{\Delta t}$
  - The time interval is equal to
- $\Delta t = t_2 - t_1$
- This is because  $t_1 - t_2$  would be a negative number
- In this interval, the momentum changes from  $p_1$  to  $p_2$ 
  - So,  $\Delta p = p_1 - p_2$
- This quantity will always be positive, since the minus sign on  $p_2$  will change a negative number into a positive number
- $Average\ force = F = \frac{\Delta p}{\Delta t} = \frac{p_1 - p_2}{t_2 - t_1}$



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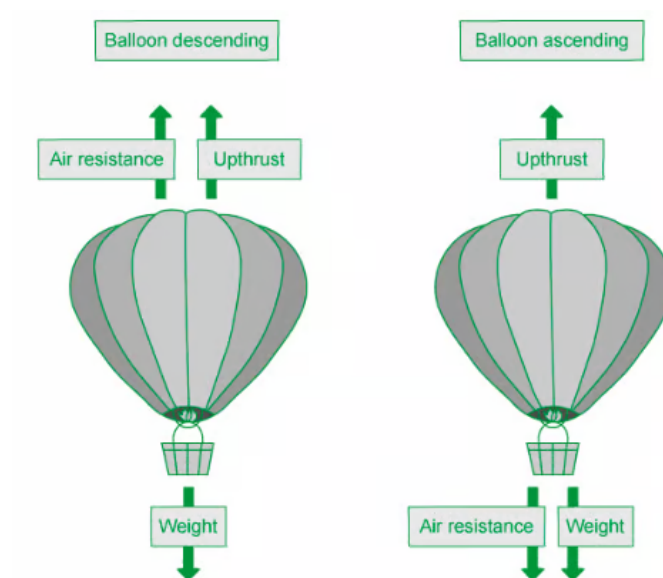
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7

The correct answer is **C** because:

- As the balloon ascends, the forces acting are as follows:
  - The weight  $W$  always acts downwards and always has a constant value
  - The upthrust force  $U$  acts upwards and changes depending on the value of the height between the top and bottom of the air balloon
  - Air resistance  $F$  opposes the direction of motion, so if the balloon is ascending, the air resistance acts downwards and changes depending on the speed
- When the balloon is descending at a constant speed, air resistance =  $F$
- A constant speed means the acceleration, and hence the resultant force = 0
- In other words, the sum of the upward forces = sum of the downward forces
  - $W = F + U$  **(1)**

- When the balloon is ascending at the same constant speed, air resistance =  $F$  (but downwards this time) and a weight  $X$  is released
  - $W + F - X = U$  (2)
- Substituting (1) into (2)
  - $F + U + F - X = U$
  - $2F - X = 0$
  - $X = 2F$



8

The correct answer is **C** because:

- Momentum is equal to
  - $\Delta p = m \Delta v$
- Force is related to momentum by
  - $F = \frac{\Delta p}{\Delta t}$
- Combining these leads to
  - $F = \frac{m \Delta v}{\Delta t}$
- Rearranging for  $\Delta t$ 
  - $m \Delta v$

$$\begin{aligned} \circ \Delta t &= \frac{F}{\dots} \\ \circ \Delta t &= \frac{(6.7 \times 10^7)(18.9 - 0)}{880000} \\ \circ \Delta t &= 1439 \text{ s} \end{aligned}$$

- Divide by 60 to convert to minutes

$$\circ \Delta t = \frac{1439}{60} = 24 \text{ minutes}$$

9

The correct answer is **C** because:

- Force is equal to the rate of change of momentum and therefore, it is directly proportional to the change in velocity  $F \propto \Delta v$
- Before the ball hits the table there is no force between the ball and table, so the force is initially zero
- When the ball comes into contact with the table its speed temporarily reduces to zero, this means there is a change in momentum, so  $F$  increases, which is shown by the initial positive gradient on the graph
- When the speed is zero, it is no longer changing, so the change in momentum, and hence force, reach a maximum which is represented on the graph by the maximum point on the curve
- Then, the speed of the ball increases in the opposite direction as it begins to bounce away from the table (but the force acts only while the ball is still in contact with the table)
- This increase in speed causes a change in momentum, and a force, but in the opposite direction, so as the ball moves away from the table, the force exerted by it gradually decreases, as seen on graph C
- At the moment the ball is about to leave the surface of the table, it has the same maximum speed as before and since the velocity/momentum is no longer changing, the force reduces to zero

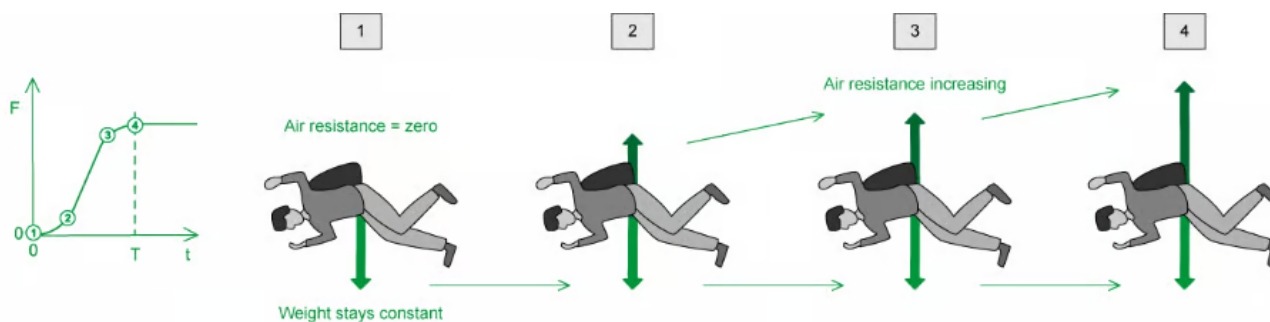
**A** is incorrect because this graph implies that the force continues to increase, even after the ball leaves the table, which is not possible

**B & D** are incorrect because there is no force between the ball and table before the ball comes into contact with it, hence the force is initially zero

10

The correct answer is **B** because:

- The best way to go about this question is to visually identify how the forces are changing as the skydiver falls and compare this to sections of the graph
- At **(1)**, the speed increases from zero and the only force acting is the skydiver's weight so she begins to accelerate
- In the first few seconds, **(1) – (2)**, the increase in air resistance is relatively small, this is because her speed is initially small and air resistance increases with speed
- Between **(2) – (3)**, the skydiver continues to accelerate due to gravity, so as the speed increases the air resistance increases too
- At **(4)**, the skydiver reaches terminal velocity, meaning that the resistance and the weight are equal and there is no longer a resultant force, hence the acceleration falls to zero and the speed remains constant (until she opens her parachute)



**A** is incorrect because the rate of increase of air resistance is not constant so the gradient of the graph cannot be constant

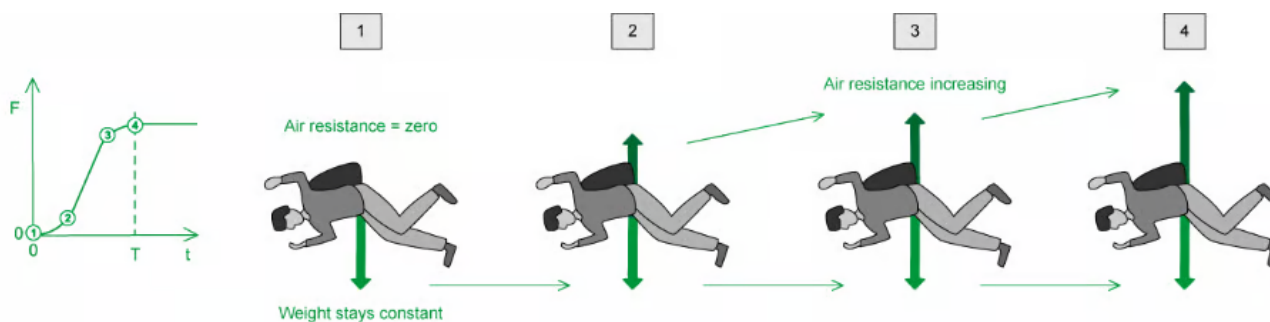
**C** is incorrect because the air resistance cannot keep increasing after time  $T$  as this would imply it becomes larger than weight so the resultant force would increase upwards

**D** is incorrect because at time  $t = 0$ , the skydiver is stationary so both the speed and air resistance start from zero



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